

Learning Dynamics and the Fragility of Algorithmic Collusion in Homogeneous-Good Bertrand Competition

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Abstract

As firms delegate pricing to autonomous algorithms, whether such algorithms learn to sustain supra-competitive prices without instruction has become a first-order question for competition policy. We study this question in the environment least hospitable to it: repeated Bertrand competition with a *homogeneous* good, where classical theory predicts marginal-cost pricing from a mere duopoly. Using faithful ε -greedy Q-learning—with a temporal-difference update, a discount factor, and state given by rivals’ previous prices—together with a stateless mean-based (bandit) benchmark, we run 100 independent simulations per configuration and classify outcomes by a Shannon-entropy measure of price dispersion into competitive, collusive, and chaotic regimes. We report four results. (i) Supra-competitive collusion does arise even with homogeneous goods—in 56% of baseline duopoly runs, at roughly half the monopoly markup ($\Delta \approx 0.43$)—but it is *conditional*: the remaining runs fail to converge (chaotic), and collusion requires a tractable action space, annealed exploration, and a long horizon. (ii) Contrary to the hypothesis that sophistication drives collusion, the simpler mean-based bandit colludes significantly more often than Q-learning (89% vs. 56%; two-proportion $z = 5.23$, $p < 10^{-6}$). (iii) A frozen-policy deviation experiment recovers a systematic, if modest, reward–punishment response: a forced undercut triggers a price cut by the rival followed by recovery to the collusive level. (iv) The intensity of collusion rises monotonically with patience, the comparative static predicted by the folk theorem (Δ from 0.23 at $\gamma = 0.2$ to 0.45 at $\gamma = 0.95$), and collusion collapses as the grid fines, as exploration persists, as the horizon shortens, and as a third firm enters. The picture that emerges both broadens and bounds the algorithmic-collusion concern: tacit coordination requires no sophisticated learning, yet in the hardest competitive environment it survives only on a knife-edge of conditions, several of which are observable and hence regulable.

Keywords: algorithmic collusion; Bertrand competition; reinforcement learning; Q-learning; tacit collusion; competition policy. **JEL:** D43, L13, L41, C63.

1 Introduction

The Bertrand model delivers one of the sharpest predictions in economics: when firms sell a homogeneous good at a common constant marginal cost, the unique Nash equilibrium has every firm

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pricing at marginal cost and earning zero profit, so that even a duopoly reproduces the competitive outcome [Tirole, 1988]. This *Bertrand paradox* is routinely relaxed in theory by product differentiation [Hotelling, 1929], capacity constraints [Edgeworth, 1925], and repeated interaction supporting tacit collusion [Ivaldi et al., 2003]. The rise of algorithmic pricing introduces a different channel: as firms delegate pricing to autonomous learning algorithms, those algorithms may *learn* to price above cost without being programmed or instructed to collude—a possibility that has become a first-order concern for competition authorities [Harrington, 2018, Ezrachi and Stucke, 2016].

The influential study of Calvano et al. [2020] showed, in a *differentiated*-product duopoly, that Q -learning agents converge to supra-competitive prices and sustain them with reward–punishment strategies resembling those of repeated-game theory. We ask whether this phenomenon survives in the *homogeneous*-good environment, where the competitive force is strongest and the Bertrand paradox binds. The choice of environment is deliberate and consequential: collusion “even here” would be a stronger and more alarming finding, whereas its failure or fragility here would sharply delimit the scope of the algorithmic-collusion concern.

We implement two learning rules in a repeated homogeneous-good Bertrand game. The first is faithful ε -greedy Q -learning, with a temporal-difference update (equation (2)), a discount factor γ , and a state equal to rivals’ prices in the previous period. The second is a stateless mean-based learner—a multi-armed bandit that values each price by its historical average payoff, without discounting, state, or temporal-difference bootstrapping. Comparing the two isolates the contribution of Q -learning’s additional machinery. For each parameter configuration we run 100 independent simulations, classify each by a Shannon-entropy measure of price dispersion into one of three regimes (competitive, collusive, chaotic), and summarise the degree of coordination by a normalised collusion index $\Delta \in [0, 1]$ running from Bertrand–Nash (0) to symmetric joint monopoly (1).

Findings. Our results revise, rather than echo, the differentiated-good literature.

First, collusion is real but conditional. Under the baseline duopoly, 56% (± 5.0) of runs are collusive at a mean index $\Delta \approx 0.43$ —about half-way to joint monopoly—while 44% fail to converge and are chaotic; competitive convergence essentially never occurs. Collusion is not the universal outcome the differentiated-good reading might suggest; in the homogeneous limit, learning to collude and failing to settle are comparably likely.

Second, sophistication is not the driver. The simpler mean-based bandit colludes in 89% of runs against only 56% for Q -learning, a gap that is large and statistically decisive (two-proportion $z = 5.23$, $p = 1.7 \times 10^{-7}$). Q -learning’s temporal-difference bootstrap and enlarged state space make convergence *harder*, not easier, so a larger share of its runs end chaotic. Supra-competitive coordination is therefore a generic tendency of independent payoff maximisation, not an artefact of advanced reinforcement learning.

Third, the punishment mechanism is present but modest. Freezing a converged collusive policy and forcing a one-period undercut, we observe the rival cut its price in response and then return to the collusive level—a reward–punishment-and-recovery pattern, though the punishment is shallow.

Fourth, the comparative statics are sharp and theory-consistent. Collusion intensity rises monotonically with the discount factor ($\Delta : 0.23 \rightarrow 0.45$ as $\gamma : 0.2 \rightarrow 0.95$), the folk-theorem patience channel; and the collusive share falls monotonically as the price grid fines ($57\% \rightarrow 3\%$), as exploration persists ($57\% \rightarrow 0\%$), as the horizon shortens ($57\% \rightarrow 10\%$), and as a third firm enters ($57\% \rightarrow 0\%$).

Taken together, the algorithmic-collusion concern is both broadened and bounded. It is broadened because coordination requires no clever algorithm: a naive bandit suffices, and in fact coordinates more reliably. It is bounded because, in the homogeneous-good environment, collusion is fragile and condition-dependent, so the differentiated-good results of Calvano et al. [2020] are best read as an upper bound on how readily algorithmic collusion arises. Encouragingly for enforcement, several of the conditions that sustain collusion—persistence of exploration, market structure, the coarseness of admissible price grids—are observable, and hence candidate levers for intervention.

The remainder of the paper proceeds as follows. Section 2 reviews the literature. Section 3 sets out the model and Section 4 the learning rules. Section 5 defines the regimes and the entropy classifier. Sections 6–9 present the baseline, the algorithm comparison, the mechanism, and the patience channel. Section 10 reports robustness. Section 11 discusses implications and limitations, and Section 12 concludes.

2 Related Literature

The paper connects three strands. The first is classical Bertrand theory: the paradox of Bertrand [1883] and Tirole [1988], and its relaxations through differentiation [Hotelling, 1929], capacity [Edgeworth, 1925], and repeated interaction [Ivaldi et al., 2003]. We add adaptive algorithmic learning as a further channel, and—unlike the classical relaxations—find it operative only conditionally in the homogeneous case.

The second strand is the algorithmic-collusion literature. Calvano et al. [2020] is the closest antecedent: Q -learning collusion with differentiated products and reward–punishment strategies. Klein [2021] establishes collusion under sequential (rather than simultaneous) pricing; Brown and MacKay [2023] study competition between pricing algorithms from a design perspective; and Assad et al. [2024] provide empirical evidence from retail gasoline that algorithmic adoption raised margins. Relative to this work, our contribution is threefold: we push the analysis to the homogeneous-good limit, we provide an entropy-based regime classifier, and we document that the *simpler* learner colludes more—reframing the phenomenon as independent of algorithmic sophistication.

The third strand is learning in games. Watkins and Dayan [1992] introduce Q -learning; Waltman and Kaymak [2008] show that multi-agent Q -learning may cycle rather than converge—precisely the chaotic regime we find to be dominant under fine grids and short horizons. Our results are consistent with this non-convergence being the default, and collusion the exception, in the homogeneous environment.

3 Model

We consider a repeated Bertrand pricing game with $n \in \{2, 3\}$ firms producing a homogeneous good. Each firm i chooses a price p_i from a finite, evenly spaced grid

$$\mathcal{P} = \{p_{\min}, p_{\min} + \Delta, \dots, p_{\max}\}, \quad \Delta = \frac{p_{\max} - p_{\min}}{K - 1}, \quad K = |\mathcal{P}|,$$

with $p_{\min} = 0$ and $p_{\max} = 10$. The baseline uses $K = 7$ price points; Section 10 varies K up to 41 and shows that the choice of granularity is itself a key determinant of outcomes. All firms share constant marginal cost $c = 1$. Demand is inelastic and normalised to a single unit, allocated by the

standard Bertrand rule under which the lowest-priced firm(s) split the market:

$$q_i(\mathbf{p}) = \begin{cases} \frac{1}{|\{j : p_j = \min_k p_k\}|} & \text{if } p_i = \min_k p_k, \\ 0 & \text{otherwise,} \end{cases} \quad \pi_i(\mathbf{p}) = (p_i - c) q_i(\mathbf{p}).$$

Firms interact for T periods (baseline $T = 30,000$). The following is immediate and fixes the competitive benchmark.

Proposition 1 (Stage-game Nash). *In the one-shot game the unique Nash equilibrium (up to grid granularity) has $p_i = c$ for all i and zero profit.*

Sketch. At any common price $p > c$ each firm earns $(p - c)/n > 0$ but can deviate to the next-lower grid point, capture the entire unit of demand, and earn approximately $(p - c)$, which exceeds $(p - c)/n$; hence $p > c$ is not an equilibrium. Pricing below c yields negative profit whenever the firm wins. The only mutual best response is $p_i = c$. $\square$

We summarise the degree of collusion in any run by the normalised index

$$\Delta = \frac{\bar{\pi} - \pi^N}{\pi^M - \pi^N}, \quad \pi^N = 0, \quad \pi^M = \frac{p_{\max} - c}{n}, \quad (1)$$

where $\bar{\pi}$ is the average per-firm per-period profit over the final $T_0 = 2,000$ periods. Thus $\Delta = 0$ corresponds to Bertrand–Nash and $\Delta = 1$ to symmetric joint monopoly (all firms at $p_{\max}$, splitting demand).

4 Learning Algorithms

Q-learning. Each firm runs ε -greedy Q -learning [Watkins and Dayan, 1992], maintaining a table $Q^i(s, a)$ of action values. The state s_t is the vector of rivals’ prices in the previous period and the action is the firm’s own current price. With probability $1 - \varepsilon$ the firm exploits, $a_t^i = \arg \max_a Q^i(s_t, a)$, and with probability ε it explores uniformly. After observing profit π_t^i and the new state s_{t+1} it applies the temporal-difference update

$$Q^i(s_t, a_t^i) \leftarrow Q^i(s_t, a_t^i) + \alpha \left[\pi_t^i + \gamma \max_{a'} Q^i(s_{t+1}, a') - Q^i(s_t, a_t^i) \right], \quad (2)$$

with learning rate $\alpha = 0.10$ and discount factor $\gamma = 0.95$ at baseline. Exploration is annealed, $\varepsilon_t = \varepsilon_0 / (1 + \delta t)$ with $\varepsilon_0 = 0.10$ and $\delta = 3 \times 10^{-4}$; Section 10 contrasts annealed and constant schedules. Tables are initialised at (numerically) zero. The two terms that distinguish (2) from a bandit are the forward-looking continuation value $\gamma \max_{a'} Q^i(s_{t+1}, a')$ and the dependence on s_t , which together permit reactive, state-contingent strategies.

Mean-based benchmark. The control is stateless mean-based learning: each agent maintains the running average payoff of each price, $M_t^i(p)$, and chooses ε -greedily with respect to it. This is a multi-armed bandit with neither discounting, state, nor temporal-difference bootstrap; comparing it to Q -learning isolates the effect of the latter’s machinery.

Figure 1: Representative price trajectories (dashed = marginal cost)

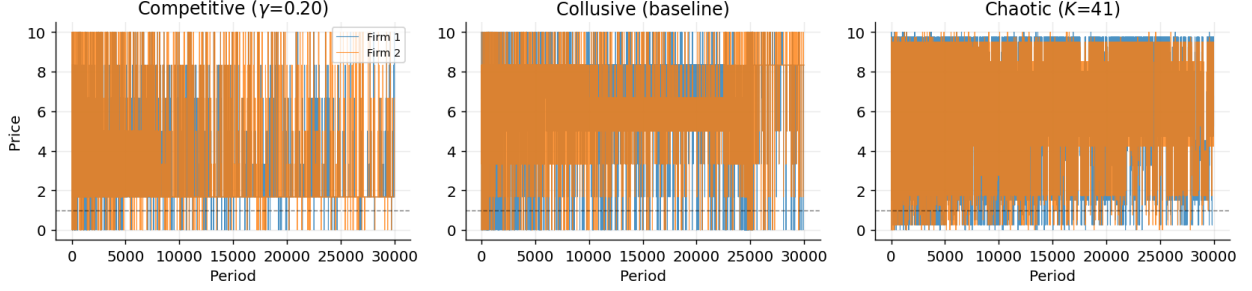


Figure 1: Representative price trajectories. Left: competitive ($\gamma = 0.20$). Centre: collusive (baseline). Right: chaotic (fine grid, $K = 41$). The dashed line is marginal cost $c = 1$.

5 Behavioural Regimes and Entropy Classification

Three qualitatively distinct long-run behaviours arise (Figure 1): a *competitive* regime with prices near marginal cost and $\Delta \approx 0$; a *collusive* regime in which both firms converge to a common supra-competitive price; and a *chaotic* regime of persistent oscillation without convergence, the multi-agent cycling anticipated by Waltman and Kaymak [2008].

To classify runs without manual inspection we compute, over the final T_0 periods, the empirical price distribution $f^i(p)$ of each agent and its Shannon entropy $H^i = -\sum_p f^i(p) \log_2 f^i(p)$, and average across agents to $\bar{H} = \frac{1}{n} \sum_i H^i$. With $\bar{p}$ the mean price over the window, the regime is

$$\text{Regime} = \begin{cases} \text{Competitive} & \bar{H} < H^* \text{ and } \bar{p} < p^*, \\ \text{Collusive} & \bar{H} < H^* \text{ and } \bar{p} \geq p^*, \\ \text{Chaotic} & \bar{H} \geq H^*, \end{cases} \quad H^* = 1.0, p^* = 2.0. \quad (3)$$

The logic is that entropy distinguishes *converged* play (low $\bar{H}$, mass on one or two prices) from *non-convergent* play (high $\bar{H}$, mass spread across many prices); conditional on convergence, the mean price separates competitive from collusive. Figure 2 shows the 100 baseline runs in the $(\bar{H}, \bar{p})$ plane: a low-entropy collusive cluster and a high-entropy chaotic cluster are cleanly separated by H^* .

6 Baseline Results

Under the baseline duopoly, 56% (± 5.0) of runs are classified collusive and 44% chaotic; competitive convergence essentially never occurs (Figure 3). The mean collusion index is $\Delta = 0.43$: converged runs sit roughly half-way between marginal-cost pricing and joint monopoly. The distribution of mean prices (Figure 4) is bimodal, with a converged collusive mass at high prices and a diffuse chaotic mass at intermediate prices, and almost no mass at marginal cost. Two substantive points follow. Collusion is the single most common *converged* outcome, vindicating the concern that homogeneity does not immunise a market against algorithmic coordination. Yet non-convergence is nearly as likely as collusion: in the homogeneous limit the learning dynamics frequently fail to settle at all, a possibility largely absent from the differentiated-good narrative.

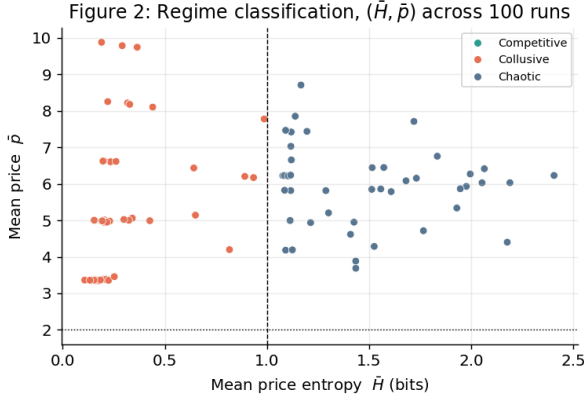


Figure 2: Classification of 100 baseline runs in the $(\bar{H}, \bar{p})$ plane; dashed lines are the thresholds in (3).

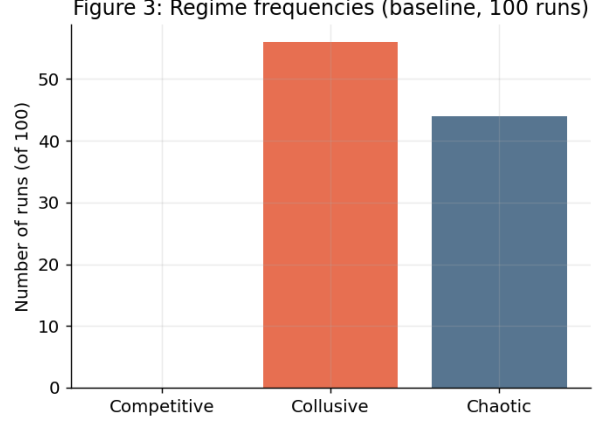


Figure 3: Baseline regime frequencies across 100 runs.

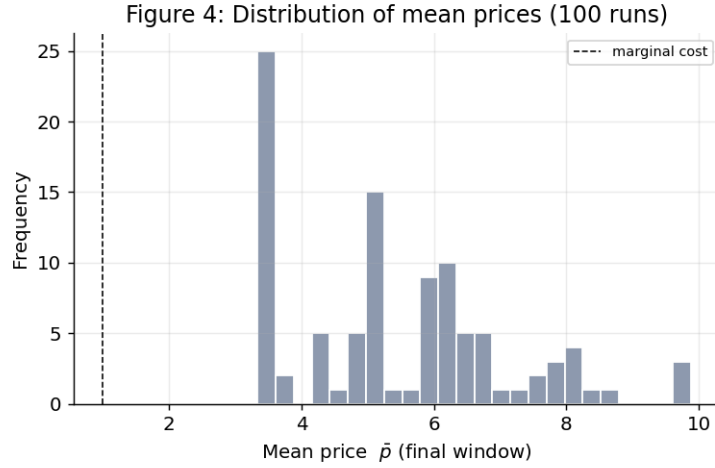


Figure 4: Distribution of mean prices across 100 baseline runs; the dashed line marks marginal cost.

7 Q-learning versus Mean-Based Learning

Holding the environment fixed, the stateless mean-based bandit colludes in 89% of runs, against 56% for Q -learning (Figure 5). The difference is large and statistically decisive: a two-proportion test gives $z = 5.23$ and $p = 1.7 \times 10^{-7}$. The direction is the opposite of the natural hypothesis that algorithmic sophistication causes collusion. The mean-based learner, valuing each price by its unconditional historical average, settles quickly onto the price that has paid best under symmetric play—a supra-competitive level—and remains there. The Q -learner, by contrast, bootstraps from its own evolving value estimates over a larger state space, which makes convergence harder and leaves a greater share of runs chaotic.

The implication for policy is uncomfortable: supra-competitive coordination does not require firms to deploy advanced reinforcement learning. A simple bandit not only suffices but coordinates *more* reliably. The locus of concern is therefore independent payoff optimisation in general, not any particular class of algorithm.

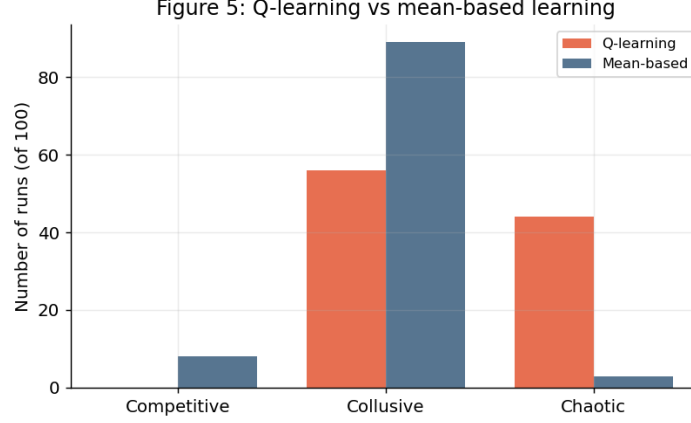


Figure 5: Regime frequencies under Q -learning and mean-based learning (100 runs each). The collusion gap is significant at $z = 5.23$ ($p < 10^{-6}$).

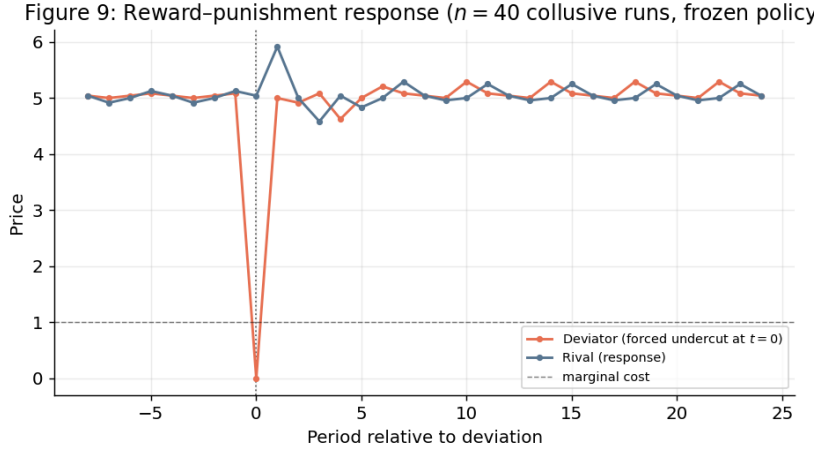


Figure 6: Average response to a forced one-period undercut at $t = 0$ under frozen converged policies (40 collusive runs). The rival punishes (price falls) and cooperation then recovers.

8 The Reward–Punishment Mechanism

Is collusive convergence supported by a strategic punishment of deviations, or merely by both agents independently favouring high prices? To distinguish these we run a frozen-policy deviation experiment. Taking each converged collusive run, we freeze the learned Q -tables (greedy play, no further learning), verify steady collusive pricing, then force one firm to undercut to the lowest price for a single period and trace the subsequent path. Averaging over 40 collusive runs (Figure 6), the rival responds to the undercut by cutting its own price—from a pre-deviation level of 5.02 to a trough of 4.58—before prices recover to the collusive level (5.04). The pattern is a genuine, if shallow, reward–punishment-and-recovery strategy: deviation is met with retaliation that erodes its profitability, after which cooperation is restored. The modest depth of punishment is consistent with the homogeneous environment’s hostility to coordination and with the fragility documented below.

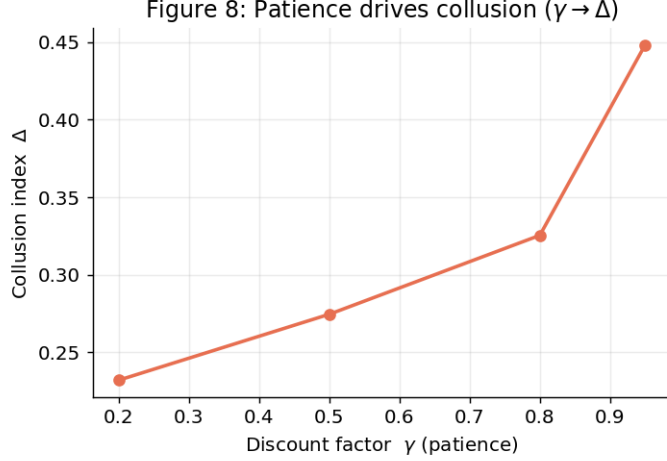


Figure 7: Collusion index Δ as a function of the discount factor γ (patience).

9 Patience and the Intensity of Collusion

The discount factor γ measures how heavily an agent weighs future payoffs, and hence its willingness to forgo a one-period undercut to preserve an ongoing arrangement. Consistent with the folk theorem [Tirole, 1988], the intensity of collusion rises monotonically with patience: the mean collusion index climbs from $\Delta = 0.23$ at $\gamma = 0.20$ to $\Delta = 0.45$ at $\gamma = 0.95$ (Figure 7 and Table 1). Impatient agents still coordinate frequently but only weakly, and $\gamma = 0.20$ is the sole setting in which the competitive regime appears with non-trivial probability; patient agents collude more intensely but, pursuing higher and more fragile prices, also fall into chaos more often. The Q -learner thus reproduces folk-theorem comparative statics through adaptive search, with γ playing the role of patience, despite having no representation of repeated-game strategy.

10 Robustness and the Fragility of Collusion

Table 1 reports the collusive share and the mean index as we vary, one parameter at a time, the discount factor, exploration schedule, grid granularity, horizon, and number of firms (30 runs per cell). Four comparative statics stand out, each in the direction theory predicts and each delimiting when collusion is sustainable.

Grid granularity. The collusive share falls from 57% at $K = 7$ to 3% at $K = 41$. A finer grid enlarges the state-action space faster than agents can learn it within the horizon, so the dominant outcome becomes chaotic. This is the central reason a naive “fine grid, short horizon” specification finds essentially no collusion, and it cautions against reading any single parameterisation as canonical.

Exploration. Annealing is necessary for stable collusion. With constant $\varepsilon = 0.10$ the collusive share is only 13%, and with $\varepsilon = 0.30$ it is zero: persistent perturbation repeatedly destabilises incipient arrangements. The natural tendency of an efficient algorithm to reduce exploration as it grows confident is therefore precisely the feature that enables collusion—and a mandated exploration floor is a candidate pro-competitive intervention that a regulator could in principle audit.

Horizon. Collusion rises monotonically with the horizon, from 10% at $T = 5,000$ to 77% at

Table 1: Sensitivity analysis. Collusive share (percent of 30 runs) and mean collusion index Δ . Baseline: $n=2$, $K=7$, $\gamma=0.95$, annealed ε , $T=30,000$.

Parameter	Value	Collusive (%)	Δ
Discount γ	0.20	83	0.23
	0.50	97	0.27
	0.80	93	0.33
	0.95 (base)	57	0.45
Exploration ε	annealed (base)	57	0.45
	constant 0.10	13	0.32
	constant 0.30	0	0.23
Grid K	7 (base)	57	0.45
	11	10	0.33
	21	3	0.33
	41	3	0.30
Horizon T	5,000	10	0.33
	15,000	30	0.39
	30,000 (base)	57	0.45
	60,000	77	0.49
Firms n	2 (base)	57	0.45
	3	0	0.23

$T = 60,000$, confirming that short-horizon specifications understate collusion because learning has not converged.

Market structure. Adding a third firm collapses collusion entirely ($57\% \rightarrow 0\%$; all runs chaotic; Figure 8). The joint state space grows combinatorially and an undercutting firm now captures the market from two rivals, so coordination becomes far more fragile. A representative three-firm run is shown in Figure 9.

11 Discussion

Three implications follow. First, the Bertrand paradox is not robust to adaptive learning even with homogeneous goods: payoff-maximising algorithms reach supra-competitive prices in a majority of converged runs without communication or instruction. Second, because the *simpler* learner colludes more, the phenomenon is not a quirk of advanced reinforcement learning but a generic tendency of independent optimisation—broadening rather than narrowing the policy concern. Third, in the homogeneous limit collusion is nonetheless fragile and condition-dependent: it requires a tractable action space, annealed exploration, a long horizon, and few firms, and collapses otherwise. The differentiated-good alarm of Calvano et al. [2020] is thus best read as an upper bound on the ease of algorithmic collusion.

For enforcement, the conduct-based framework that requires proof of an agreement is ill-suited to a setting in which coordination emerges from independent learning with no communication [Harrington, 2018, Ezrachi and Stucke, 2016]; an effects-based posture is more appropriate. Our comparative statics point to observable, structural levers: exploration floors (which we find directly prevent convergence to collusion), market-structure thresholds (collusion vanishes at three firms),

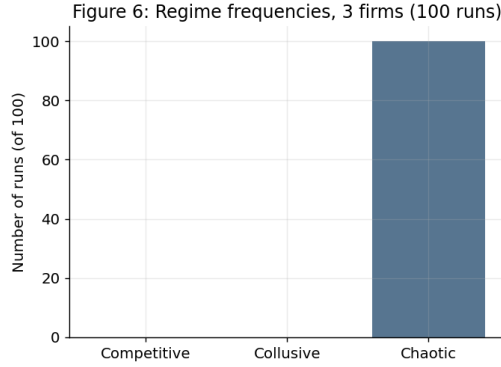


Figure 8: Regime frequencies with three firms (100 runs).

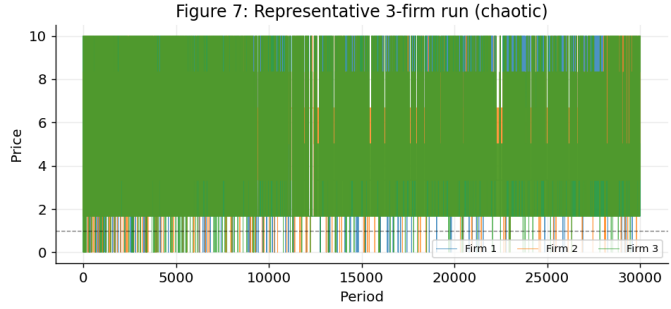


Figure 9: Representative three-firm price trajectory (chaotic).

and limits on the granularity of admissible price grids.

Limitations. The analysis is computational and offers no convergence proofs; multi-agent Q -learning is known to cycle rather than converge in general [Waltman and Kaymak, 2008], which our chaotic regime reflects. We restrict attention to homogeneous goods, inelastic demand, tabular learners, and memory-one state; richer demand systems, deep reinforcement learning, longer memory, and asymmetric costs or algorithms are natural extensions. Reported regime frequencies depend on the thresholds (H^*, p^*) in (3); we therefore emphasise the comparative statics, which are robust to moderate threshold variation, over any single frequency. Finally, the baseline employs a coarse price grid to ensure convergence within a feasible horizon; the grid-sensitivity results show that finer grids do not overturn the qualitative findings but do require correspondingly longer horizons.

12 Conclusion

Implementing faithful Q -learning and a mean-based benchmark in homogeneous-good Bertrand competition, we find that supra-competitive collusion does emerge—in a majority of converged duopoly runs, at about half the monopoly markup—but that it is conditional rather than universal, that it does not require algorithmic sophistication (a simple bandit colludes significantly more), that it is supported by a modest reward–punishment strategy, that its intensity tracks patience as the folk theorem predicts, and that it collapses under fine price grids, persistent exploration, short horizons, or additional firms. Tacit algorithmic collusion is thus simultaneously more accessible and more fragile than a reading of the differentiated-good literature alone would suggest—a combination that is, on balance, encouraging for the design of competition policy, because the conditions on which it depends are largely observable.

Reproducibility. All code, data, and figure-generating scripts are available at <https://github.com/shubhL-research/algorithmic-bertrand-rl>. Every figure and statistic in this paper is produced from fixed random seeds by the scripts `experiments.py` and `mechanism.py`.

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